E13: The Band-Pass Amplitude Equation as a Negative Gradient Flow

A Complete Variational and Dissipation Proof

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Abstract

We prove that the complex band-pass amplitude equation $\partial_t A = (r - a\Delta - b\Delta^2)A - \beta|A|^2 A$ is the negative Wirtinger gradient flow of the associated Swift–Hohenberg energy. We compute the first variation, establish an exact energy–dissipation identity, identify the normalization relative to the real L^2 gradient, and recover the preferred Fourier shell. The claim is made under explicit periodic boundary and regularity assumptions; it does not imply global well-posedness or convergence to a unique pattern.

1 Setting and statement

Let $\Omega = \mathbb{T}^d$ be a flat periodic torus, with $1 \le d \le 3$, and let $r \in \mathbb{R}$, $a, b > 0$, and $\beta \in \mathbb{R}$. Complex L^2 is viewed as a real Hilbert space with inner product

$$(u, v)_{\mathbb{R}} = \operatorname{Re} \int_\Omega u\overline{v}\, \mathrm{d}x. \tag{1}$$

Define the real-valued energy for $A \in H^2(\Omega; \mathbb{C})$ by

$$\mathcal{E}[A] = \int_\Omega \left(-r|A|^2 - a|\nabla A|^2 + b|\Delta A|^2 + \frac{\beta}{2}|A|^4 \right) \mathrm{d}x. \tag{2}$$

This energy is well defined for $d \le 3$ because $H^2(\Omega)$ embeds continuously into $L^\infty(\Omega)$.

Theorem 1 (E13 negative-gradient-flow identity)**.** *Suppose*

$$A \in C^1([0, T]; H^2(\Omega; \mathbb{C})) \cap C([0, T]; H^4(\Omega; \mathbb{C})) \tag{3}$$

and A solves, as an identity in $L^2(\Omega)$,

$$\partial_t A = (r - a\Delta - b\Delta^2)A - \beta|A|^2 A. \tag{4}$$

Then the Wirtinger variational derivative (with the convention below) is

$$\frac{\delta \mathcal{E}}{\delta \overline{A}} = -rA + a\Delta A + b\Delta^2 A + \beta|A|^2 A, \tag{5}$$

so that

$$\partial_t A = -\frac{\delta \mathcal{E}}{\delta \overline{A}}. \tag{6}$$

Moreover, for every $t \in [0,T]$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}[A(t)] = -2\|\partial_t A(t)\|^2_{L^2(\Omega)} \le 0, \tag{7}$$

and consequently

$$\mathcal{E}[A(T)] + 2\int_0^T \|\partial_t A(t)\|^2_{L^2(\Omega)}\,\mathrm{d}t = \mathcal{E}[A(0)]. \tag{8}$$

Equality in (7) at time t holds if and only if $\partial_t A(t) = 0$ in $L^2(\Omega)$, equivalently if $A(t)$ satisfies

$$(r - a\Delta - b\Delta^2)A - \beta|A|^2 A = 0. \tag{9}$$

2 First variation: proof of the gradient formula

Proof of Theorem 1. Fix $A \in H^4(\Omega;\mathbb{C})$ and a smooth periodic test function $h \in C^\infty(\Omega;\mathbb{C})$. For real ε, put $A_\varepsilon = A + \varepsilon h$. We use the convention

$$D\mathcal{E}[A](h) = 2\,\mathrm{Re}\int_\Omega \frac{\delta\mathcal{E}}{\delta\overline{A}}\,\overline{h}\,\mathrm{d}x. \tag{10}$$

This is the standard conjugate-variable/Wirtinger normalization for a real-valued functional.

Quadratic amplitude term. Since

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}|A+\varepsilon h|^2\Big|_{\varepsilon=0} = 2\,\mathrm{Re}(A\overline{h}),$$

the contribution of $-r\int|A|^2$ is

$$-2r\,\mathrm{Re}\int_\Omega A\overline{h}\,\mathrm{d}x. \tag{11}$$

Gradient term. Differentiating $-a\int|\nabla A|^2$ gives

$$-2a\,\mathrm{Re}\int_\Omega \nabla A\cdot\nabla\overline{h}\,\mathrm{d}x.$$

By periodic integration by parts, $\int_\Omega \nabla A\cdot\nabla\overline{h} = -\int_\Omega(\Delta A)\overline{h}$. Thus this contribution is

$$2a\,\mathrm{Re}\int_\Omega(\Delta A)\overline{h}\,\mathrm{d}x. \tag{12}$$

Fourth-order term. Differentiating $b\int|\Delta A|^2$ yields

$$2b\,\mathrm{Re}\int_\Omega(\Delta A)(\Delta\overline{h})\,\mathrm{d}x.$$

Two periodic integrations by parts, or self-adjointness of Δ, give $\int(\Delta A)(\Delta\overline{h}) = \int(\Delta^2 A)\overline{h}$. Therefore the contribution is

$$2b\,\mathrm{Re}\int_\Omega(\Delta^2 A)\overline{h}\,\mathrm{d}x. \tag{13}$$

Quartic term. The pointwise identity

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}|A+\varepsilon h|^4\Big|_{\varepsilon=0} = 4|A|^2\,\mathrm{Re}(A\overline{h})$$

shows that the contribution of $(\beta/2)\int|A|^4$ is

$$2\beta\,\mathrm{Re}\int_\Omega |A|^2 A\overline{h}\,\mathrm{d}x. \tag{14}$$

Combination. Adding (11)–(14) gives

$$D\mathcal{E}[A](h) = 2\,\mathrm{Re}\int_\Omega(-rA + a\Delta A + b\Delta^2 A + \beta|A|^2A)\overline{h}\,\mathrm{d}x. \tag{15}$$

By density, (15) holds for all $h\in H^2(\Omega)$. Comparison with (10) proves (5). Taking its negative gives exactly the right-hand side of (4), establishing (6).

Energy dissipation. The regularity (3) justifies the H^2 chain rule:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}[A(t)] = D\mathcal{E}[A(t)](\partial_t A(t)).$$

Substituting $h=\partial_t A$ into (15) and then using (6),

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}[A(t)] &= 2\,\mathrm{Re}\int_\Omega \frac{\delta\mathcal{E}}{\delta\overline{A}}\overline{\partial_t A}\,\mathrm{d}x\\ &= -2\,\mathrm{Re}\int_\Omega \partial_t A\,\overline{\partial_t A}\,\mathrm{d}x\\ &= -2\|\partial_t A\|^2_{L^2(\Omega)}.\end{aligned}$$

This proves (7). Integrating in time proves (8). Finally, the integrand is a squared norm, so it vanishes exactly when $\partial_t A=0$, which is equivalent to (9) via (4). □

3 Normalization of the real L^2 gradient

Because the real L^2 gradient is defined by

$$D\mathcal{E}[A](h) = (\nabla_{\mathbb{R}}\mathcal{E}[A], h)_{\mathbb{R}},$$

comparison with (15) implies

$$\nabla_{\mathbb{R}}\mathcal{E}[A] = 2\frac{\delta\mathcal{E}}{\delta\overline{A}}. \tag{16}$$

Consequently the equation can also be written

$$\partial_t A = -\tfrac{1}{2}\nabla_{\mathbb{R}}\mathcal{E}[A].$$

If $\mathcal{F}[A]=\mathcal{E}[A]/2$, then E13 is a literal negative real L^2 gradient flow:

$$\partial_t A = -\nabla_{\mathbb{R}}\mathcal{F}[A], \qquad \frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}[A(t)] = -\|\partial_t A(t)\|^2_{L^2(\Omega)}. \tag{17}$$

The factor of two is a convention and does not change the dynamics.

4 Centered Swift–Hohenberg representation

Define

$$k_\star^2 = \frac{a}{2b}, \qquad \mu = r + \frac{a^2}{4b}. \tag{18}$$

Direct expansion shows

$$r - a\Delta - b\Delta^2 = \mu - b(\Delta + k_\star^2)^2,$$

so the same equation takes the centered form

$$\partial_t A = \mu A - b(\Delta + k_\star^2)^2 A - \beta |A|^2 A. \tag{19}$$

Consider

$$\mathcal{E}_{\mathrm{SH}}[A] = \int_\Omega \left[-\mu |A|^2 + b|(\Delta + k_\star^2)A|^2 + \frac{\beta}{2}|A|^4 \right] \mathrm{d}x. \tag{20}$$

Periodic integration by parts gives

$$\begin{aligned} \|(\Delta + k_\star^2)A\|_2^2 &= \|\Delta A\|_2^2 + 2k_\star^2 \,\mathrm{Re} \int_\Omega (\Delta A)\overline{A}\,\mathrm{d}x + k_\star^4 \|A\|_2^2 \\ &= \|\Delta A\|_2^2 - 2k_\star^2 \|\nabla A\|_2^2 + k_\star^4 \|A\|_2^2. \end{aligned}$$

Since $2bk_\star^2 = a$ and $bk_\star^4 - \mu = -r$, it follows that

$$\mathcal{E}_{\mathrm{SH}}[A] = \mathcal{E}[A] \qquad (A \in H^2(\Omega;\mathbb{C})). \tag{21}$$

Thus the centered and uncentered energy descriptions are identical, not merely equivalent up to an additive constant.

5 Preferred Fourier shell

Linearizing (4) at $A = 0$ and substituting a Fourier mode $e^{ik\cdot x}$ gives

$$\partial_t \widehat{A}_k = \lambda(k)\widehat{A}_k, \qquad \lambda(k) = r + a|k|^2 - b|k|^4.$$

Completion of the square using (18) yields

$$\lambda(k) = \mu - b(|k|^2 - k_\star^2)^2. \tag{22}$$

In the continuum variable $q = |k|^2 \geq 0$, the function $\lambda(q) = r + aq - bq^2$ has

$$\lambda'(q) = a - 2bq, \qquad \lambda''(q) = -2b < 0.$$

Its unique maximizing radius is therefore $|k| = k_\star > 0$. When $\mu > 0$, the linearly growing modes satisfy

$$||k|^2 - k_\star^2| < \sqrt{\mu/b}. \tag{23}$$

On the torus, only discrete lattice wavenumbers occur; the actual fastest available modes maximize (22) over that lattice. The nonlinear equation does not in general preserve a finite Fourier shell exactly.

6 Lower bound when $\beta > 0$

Corollary 1. *If $\beta > 0$ and Ω has finite volume, then*

$$\mathcal{E}[A] \geq -\frac{|\Omega|}{2\beta}(\max\{\mu, 0\})^2. \tag{24}$$

If a solution satisfying the theorem's regularity assumptions exists globally forward in time, then

$$\int_0^\infty \|\partial_t A(t)\|_2^2 \, \mathrm{d}t < \infty. \tag{25}$$

Proof. For $s \geq 0$, completing the square (or minimizing the quadratic on $[0, \infty)$) gives

$$\frac{\beta}{2}s^2 - \mu s \geq -\frac{(\max\{\mu, 0\})^2}{2\beta}.$$

Take $s = |A(x)|^2$ and integrate in (20). The remaining fourth-order square is nonnegative, proving (24). Integrate (7) from 0 to T, apply the lower bound, and let $T \to \infty$ to obtain (25). □

Remark 1 (Scope)**.** *The algebraic first-variation identity does not require $\beta > 0$; positivity is used only for the lower-bound corollary. The proof establishes monotone energy descent for regular solutions, but does not by itself prove global well-posedness, convergence to a unique steady state, nonlinear Fourier-shell invariance, or derivation of the amplitude equation from a more fundamental physical model. On other domains the same computation holds when boundary conditions eliminate all boundary terms and give the required self-adjoint operator.*

Conclusion

Under the hypotheses of Theorem 1,

$$\boxed{\partial_t A = -\frac{\delta \mathcal{E}}{\delta \overline{A}}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}[A(t)] = -2\|\partial_t A(t)\|_2^2 \leq 0.}$$

This is the complete E13 negative-gradient-flow and Lyapunov-dissipation identity for the specified periodic Swift–Hohenberg amplitude model.

Equation registry. https://github.com/rickyjreyes/geometry_of_resonance/blob/main/WCT_FULL_EQUATION_LIST_CORRECTED.md